POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5033D | Semester V | Theory | 4 Credits

Renewable Power Plants

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5033D as Renewable Power Plants in Semester V for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Roorkee’s Smart Grid course and polytechnic renewable power curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Power plant designs, grid-connection settings, battery management logic and synchronization protocols must be based on approved engineering plans, certified equipment specifications, current safety standards and competent professional review.

Verified source note: Verified sources support solar PV, wind generators (DFIG/PMSG), energy storage, microgrids, and grid integration issues. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Renewable Power Plants studies the engineering, operation and grid integration of non-conventional energy generation systems. It develops the ability to analyze solar photovoltaic (PV) and wind power plants, evaluate energy storage technologies and microgrid architectures, and understand grid synchronization and power quality requirements while respecting the technical and safety boundaries of modern power plant engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□.

Prerequisite foundation

Electrical machines, power electronics, power systems, basic electronics and environmental engineering.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the components, operation and grid connection of solar photovoltaic (PV) power plants.
2. Explain the working principles of wind power plants, including generator types (DFIG, PMSG) and power conversion.
3. Explain energy storage technologies, battery management systems (BMS) and the architecture of AC/DC microgrids.
4. Explain the technical requirements for grid integration, synchronization and power quality in renewable power systems.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the design and operation of solar PV power plants and MPPT control strategies.	Understand
CO2	Explain wind energy conversion systems and the role of power electronics in wind power plants.	Understand
CO3	Explain the role of energy storage and microgrid control in distributed generation.	Understand
CO4	Explain grid synchronization protocols and power quality issues in renewable energy integration.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Solar Photovoltaic (PV) Power Plants	Solar PV cell fundamentals; modules and arrays; balance of system (BOS); Maximum Power Point Tracking (MPPT) techniques; grid-connected vs stand-alone systems; solar inverters; plant layout and performance monitoring.	Approx. 16 hours	CO1	Module 1
2	Wind Power Plants and Conversion	Wind energy conversion systems (WECS); generator types (Fixed speed vs Variable speed); Doubly Fed Induction Generator (DFIG); Permanent Magnet Synchronous Generator (PMSG); power electronic interfaces; wind farm control.	Approx. 16 hours	CO2	Module 2
3	Energy Storage and Microgrids	Need for energy storage; storage technologies (Lead-acid, Li-ion, Flow batteries, Pumped hydro); Battery Management Systems (BMS); AC and DC microgrid architectures; islanded vs grid-connected operation; microgrid control.	Approx. 14 hours	CO3	Module 3
4	Grid Integration and Power Quality	Technical challenges of renewable integration; grid codes and standards; synchronization techniques; power quality issues (harmonics, flicker, voltage sags); reactive power support; future trends (smart grid, V2G).	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Solar Photovoltaic (PV) Power Plants

Solar PV cell fundamentals; modules and arrays; balance of system (BOS); Maximum Power Point Tracking (MPPT) techniques; grid-connected vs stand-alone systems; solar inverters; plant layout and performance monitoring.

CO1 · Approx. 16 hours

Wind Power Plants and Conversion

Wind energy conversion systems (WECS); generator types (Fixed speed vs Variable speed); Doubly Fed Induction Generator (DFIG); Permanent Magnet Synchronous Generator (PMSG); power electronic interfaces; wind farm control.

CO2 · Approx. 16 hours

Energy Storage and Microgrids

Need for energy storage; storage technologies (Lead-acid, Li-ion, Flow batteries, Pumped hydro); Battery Management Systems (BMS); AC and DC microgrid architectures; islanded vs grid-connected operation; microgrid control.

CO3 · Approx. 14 hours

Grid Integration and Power Quality

Technical challenges of renewable integration; grid codes and standards; synchronization techniques; power quality issues (harmonics, flicker, voltage sags); reactive power support; future trends (smart grid, V2G).

CO4 · Approx. 14 hours

MODULE 1

Solar Photovoltaic (PV) Power Plants

Solar PV cell fundamentals; modules and arrays; balance of system (BOS); Maximum Power Point Tracking (MPPT) techniques; grid-connected vs stand-alone systems; solar inverters; plant layout and performance monitoring.

Approx. 16 hours **CO1** **Understand · Apply**

Learning goals

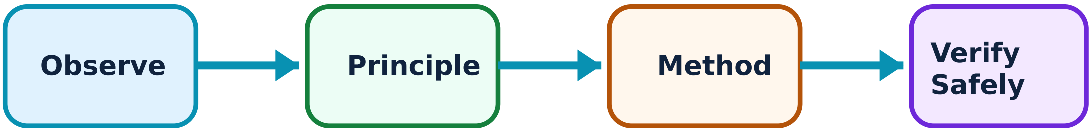
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Solar Photovoltaic (PV) Power Plants: identify the system, state its principle, follow the correct method and verify the result safely.

1. PV system components and MPPT−

Solar PV plants convert sunlight directly into electricity. The balance of system (BOS) includes inverters, mounting structures and cables. MPPT (Maximum Power Point Tracking) algorithms like Perturb and Observe (P&O) ensure the PV array operates at its peak efficiency under varying solar radiation.

Study and application method

Describe the working of a grid-connected solar inverter; explain the 'Perturb and Observe' MPPT algorithm conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the working of a grid-connected solar inverter; explain the 'Perturb and Observe' MPPT algorithm conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: PV arrays involve high DC voltages. Do not perform maintenance on live arrays without authorised safety equipment and DC isolation procedures.

2. Grid connection and monitoring−

Grid-connected systems feed power into the utility grid, requiring synchronization and anti-islanding protection. Performance monitoring systems track energy yield, performance ratio (PR) and detect faults in real-time to ensure optimal plant operation.

Study and application method

List the functions of a 'Net Meter' in a rooftop solar system; define 'Performance Ratio' (PR) for a solar power plant.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Method: List the functions of a 'Net Meter' in a rooftop solar system; define 'Performance Ratio' (PR) for a solar power plant.

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Common mistake / precaution: Improper grounding of PV structures can lead to electrical hazards. All plant installations must follow authorised electrical safety codes and grounding standards.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Wind Power Plants and Conversion

Wind energy conversion systems (WECS); generator types (Fixed speed vs Variable speed); Doubly Fed Induction Generator (DFIG); Permanent Magnet Synchronous Generator (PMSG); power electronic interfaces; wind farm control.

Approx. 16 hours | **CO2** | **Understand · Apply**

Learning goals

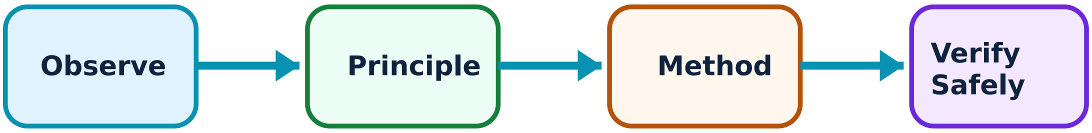
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Wind Power Plants and Conversion: identify the system, state its principle, follow the correct method and verify the result safely.

1. Wind generators and conversion−

Modern wind plants use variable speed generators like DFIG or PMSG. DFIG allows for control of active and reactive power via the rotor circuit. PMSG uses full-scale power converters, providing better grid support and higher efficiency at varying wind speeds.

Study and application method

Compare DFIG and PMSG generators in a conceptual table for wind power applications; explain the role of the 'back-to-back' converter in a wind turbine.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare DFIG and PMSG generators in a conceptual table for wind power applications; explain the role of the 'back-to-back' converter in a wind turbine.

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Common mistake / precaution: Wind turbines are complex mechanical and electrical systems. Do not operate turbines in extreme weather without authorised automatic braking and pitch control.

2. Wind farm architecture−

Wind farms consist of multiple turbines connected to a central substation. Control systems coordinate the power output, manage reactive power and ensure the farm meets grid requirements for voltage and frequency stability.

Study and application method

Sketch the layout of a typical onshore wind farm; list three factors that affect the selection of a wind turbine site.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the layout of a typical onshore wind farm; list three factors that affect the selection of a wind turbine site.

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Common mistake / precaution: Wind turbines produce flicker and noise. All plant locations must meet authorised environmental and community impact standards.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Energy Storage and Microgrids

Need for energy storage; storage technologies (Lead-acid, Li-ion, Flow batteries, Pumped hydro); Battery Management Systems (BMS); AC and DC microgrid architectures; islanded vs grid-connected operation; microgrid control.

Approx. 14 hours **CO3** **Understand · Apply**

Learning goals

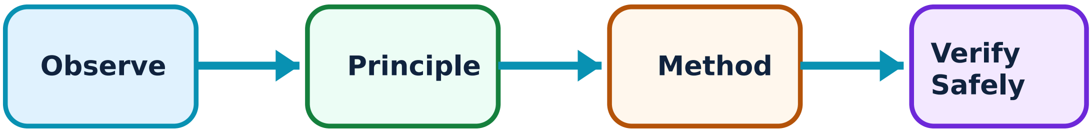
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Energy Storage and Microgrids: identify the system, state its principle, follow the correct method and verify the result safely.

1. Energy storage and BMS−

Storage mitigates the intermittency of renewables. Li-ion batteries are common for distributed storage. The Battery Management System (BMS) monitors cell voltage, temperature and state of charge (SoC) to ensure safety and longevity of the battery bank.

Study and application method

List three types of energy storage technologies and their typical applications; explain the primary functions of a 'Battery Management System'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List three types of energy storage technologies and their typical applications; explain the primary functions of a 'Battery Management System'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Batteries involve chemical and fire hazards. Do not install or maintain storage systems without authorised ventilation and fire suppression systems.

2. Microgrid architecture and control−

Microgrids are localized grids that can operate independently (islanded) or in parallel with the main grid. They integrate multiple renewable sources and storage, using a microgrid controller to balance supply and demand and manage transitions between modes.

Study and application method

Compare AC microgrids and DC microgrids in terms of efficiency and complexity; describe the concept of 'islanding' in a microgrid.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare AC microgrids and DC microgrids in terms of efficiency and complexity; describe the concept of 'islanding' in a microgrid.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Transitioning to islanded mode requires precise load shedding and frequency control. Do not island a microgrid without authorised protection and control coordination.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Grid Integration and Power Quality

Technical challenges of renewable integration; grid codes and standards; synchronization techniques; power quality issues (harmonics, flicker, voltage sags); reactive power support; future trends (smart grid, V2G).

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

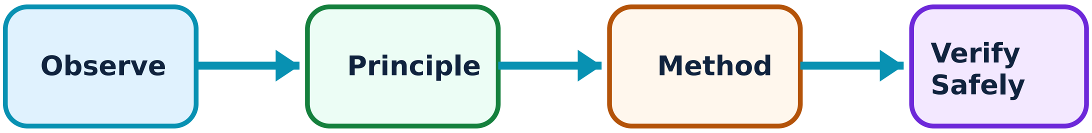
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Grid Integration and Power Quality: identify the system, state its principle, follow the correct method and verify the result safely.

1. Integration and synchronization−

Renewable integration requires adherence to grid codes (e.g., IEEE 1547). Synchronization involves matching voltage, frequency and phase. Advanced inverters provide 'grid-forming' capabilities to support grid stability during disturbances.

Study and application method

List the requirements for synchronizing a large-scale solar farm with the utility grid; define 'Low Voltage Ride Through' (LVRT).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the requirements for synchronizing a large-scale solar farm with the utility grid; define 'Low Voltage Ride Through' (LVRT).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Grid disturbances can damage power electronic converters. Ensure all integration points have authorised protective relaying and surge protection.

2. Power quality and smart grid−

Renewables can introduce harmonics and voltage fluctuations. Power quality monitors and active filters are used for mitigation. The smart grid integrates advanced communication and control to manage bidirectional power flows and electric vehicle (V2G) integration.

Study and application method

Describe how harmonics are produced by power electronic converters; explain the concept of 'Vehicle-to-Grid' (V2G) technology.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe how harmonics are produced by power electronic converters; explain the concept of 'Vehicle-to-Grid' (V2G) technology.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Smart grid systems are vulnerable to cyber threats. All communication and control networks must follow authorised cybersecurity protocols for critical infrastructure.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Solar Photovoltaic (PV) Power Plants

Use the concept flow to revise observation, principle, method and verification.

Module 2: Wind Power Plants and Conversion

Use the concept flow to revise observation, principle, method and verification.

Module 3: Energy Storage and Microgrids

Use the concept flow to revise observation, principle, method and verification.

Module 4: Grid Integration and Power Quality

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare solar PV and wind power plants in a conceptual table showing their capacity factor and typical location.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the block diagram of a DFIG-based wind energy conversion system.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the architecture of a hybrid AC/DC microgrid showing sources, storage and loads.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the energy yield conceptually for a 1MW solar plant given the daily solar insolation.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the grid-integration standards (e.g., IEEE/IEC) applicable to renewable power plants in your region.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Solar Photovoltaic (PV) Power Plants

Explain PV system components and MPPT and state one correct method, application or precaution.−

Solar PV plants convert sunlight directly into electricity. The balance of system (BOS) includes inverters, mounting structures and cables. MPPT (Maximum Power Point Tracking) algorithms like Perturb and Observe (P&O) ensure the PV array operates at its peak efficiency under varying solar radiation.

Answer framework: Describe the working of a grid-connected solar inverter; explain the 'Perturb and Observe' MPPT algorithm conceptually.

Important point: PV arrays involve high DC voltages. Do not perform maintenance on live arrays without authorised safety equipment and DC isolation procedures.

Explain Grid connection and monitoring and state one correct method, application or precaution.−

Grid-connected systems feed power into the utility grid, requiring synchronization and anti-islanding protection. Performance monitoring systems track energy yield, performance ratio (PR) and detect faults in real-time to ensure optimal plant operation.

Answer framework: List the functions of a 'Net Meter' in a rooftop solar system; define 'Performance Ratio' (PR) for a solar power plant.

Important point: Improper grounding of PV structures can lead to electrical hazards. All plant installations must follow authorised electrical safety codes and grounding standards.

Module 2 — Wind Power Plants and Conversion

Explain Wind generators and conversion and state one correct method, application or precaution.−

Modern wind plants use variable speed generators like DFIG or PMSG. DFIG allows for control of active and reactive power via the rotor circuit. PMSG uses full-scale power converters, providing better grid support and higher efficiency at varying wind speeds.

Answer framework: Compare DFIG and PMSG generators in a conceptual table for wind power applications; explain the role of the 'back-to-back' converter in a wind turbine.

Important point: Wind turbines are complex mechanical and electrical systems. Do not operate turbines in extreme weather without authorised automatic braking and pitch control.

Explain Wind farm architecture and state one correct method, application or precaution.−

Wind farms consist of multiple turbines connected to a central substation. Control systems coordinate the power output, manage reactive power and ensure the farm meets grid requirements for voltage and frequency stability.

Answer framework: Sketch the layout of a typical onshore wind farm; list three factors that affect the selection of a wind turbine site.

Important point: Wind turbines produce flicker and noise. All plant locations must meet authorised environmental and community impact standards.

Module 3 — Energy Storage and Microgrids

Explain Energy storage and BMS and state one correct method, application or precaution.−

Storage mitigates the intermittency of renewables. Li-ion batteries are common for distributed storage. The Battery Management System (BMS) monitors cell voltage, temperature and state of charge (SoC) to ensure safety and longevity of the battery bank.

Answer framework: List three types of energy storage technologies and their typical applications; explain the primary functions of a 'Battery Management System'.

Important point: Batteries involve chemical and fire hazards. Do not install or maintain storage systems without authorised ventilation and fire suppression systems.

Explain Microgrid architecture and control and state one correct method, application or precaution.−

Microgrids are localized grids that can operate independently (islanded) or in parallel with the main grid. They integrate multiple renewable sources and storage, using a microgrid controller to balance supply and demand and manage transitions between modes.

Answer framework: Compare AC microgrids and DC microgrids in terms of efficiency and complexity; describe the concept of 'islanding' in a microgrid.

Important point: Transitioning to islanded mode requires precise load shedding and frequency control. Do not island a microgrid without authorised protection and control coordination.

Module 4 — Grid Integration and Power Quality

Explain Integration and synchronization and state one correct method, application or precaution.−

Renewable integration requires adherence to grid codes (e.g., IEEE 1547). Synchronization involves matching voltage, frequency and phase. Advanced inverters provide 'grid-forming' capabilities to support grid stability during disturbances.

Answer framework: List the requirements for synchronizing a large-scale solar farm with the utility grid; define 'Low Voltage Ride Through' (LVRT).

Important point: Grid disturbances can damage power electronic converters. Ensure all integration points have authorised protective relaying and surge protection.

Explain Power quality and smart grid and state one correct method, application or

precaution.−

Renewables can introduce harmonics and voltage fluctuations. Power quality monitors and active filters are used for mitigation. The smart grid integrates advanced communication and control to manage bidirectional power flows and electric vehicle (V2G) integration.

Answer framework: Describe how harmonics are produced by power electronic converters; explain the concept of 'Vehicle-to-Grid' (V2G) technology.

Important point: Smart grid systems are vulnerable to cyber threats. All communication and control networks must follow authorised cybersecurity protocols for critical infrastructure.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Solar Photovoltaic (PV) Power Plants.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Wind Power Plants and Conversion.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Energy Storage and Microgrids.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Grid Integration and Power Quality.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Solar PV cell fundamentals; modules and arrays; balance of system (BOS); Maximum Power Point Tracking (MPPT) techniques; grid-connected vs stand-alone systems; solar inverters; plant layout and performance monitoring..

2. Develop a complete answer or practical plan for: Wind energy conversion systems (WECS); generator types (Fixed speed vs Variable speed); Doubly Fed Induction Generator (DFIG); Permanent Magnet Synchronous Generator (PMSG); power electronic interfaces; wind farm control..
3. Develop a complete answer or practical plan for: Need for energy storage; storage technologies (Lead-acid, Li-ion, Flow batteries, Pumped hydro); Battery Management Systems (BMS); AC and DC microgrid architectures; islanded vs grid-connected operation; microgrid control..
4. Develop a complete answer or practical plan for: Technical challenges of renewable integration; grid codes and standards; synchronization techniques; power quality issues (harmonics, flicker, voltage sags); reactive power support; future trends (smart grid, V2G)..

LAST-MILE REVISION

Quick Revision

Module 1: Solar Photovoltaic (PV) Power Plants

Solar PV cell fundamentals; modules and arrays; balance of system (BOS); Maximum Power Point Tracking (MPPT) techniques; grid-connected vs stand-alone systems; solar inverters; plant layout and performance monitoring.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Wind Power Plants and Conversion

Wind energy conversion systems (WECS); generator types (Fixed speed vs Variable speed); Doubly Fed Induction Generator (DFIG); Permanent Magnet Synchronous Generator (PMSG); power electronic interfaces; wind farm control.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Energy Storage and Microgrids

Need for energy storage; storage technologies (Lead-acid, Li-ion, Flow batteries, Pumped hydro); Battery Management Systems (BMS); AC and DC microgrid architectures; islanded vs grid-connected operation; microgrid control.

- Match each topic to CO3.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 4: Grid Integration and Power Quality

Technical challenges of renewable integration; grid codes and standards; synchronization techniques; power quality issues (harmonics, flicker, voltage sags); reactive power support; future trends (smart grid, V2G).

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
PV system components and MPPT	Solar PV plants convert sunlight directly into electricity.
Grid connection and monitoring	Grid-connected systems feed power into the utility grid, requiring synchronization and anti-islanding protection.
Wind generators and conversion	Modern wind plants use variable speed generators like DFIG or PMSG.
Wind farm architecture	Wind farms consist of multiple turbines connected to a central substation.
Energy storage and BMS	Storage mitigates the intermittency of renewables.
Microgrid architecture and control	Microgrids are localized grids that can operate independently (islanded) or in parallel with the main grid.
Integration and synchronization	Renewable integration requires adherence to grid codes (e.
Power quality and smart grid	Renewables can introduce harmonics and voltage fluctuations.

LEARNING RESOURCES

References

Recommended renewable-power references

1. S. N. Bhadra, D. Kastha and S. Banerjee, Wind Electrical Systems.
2. Chetan Singh Solanki, Solar Photovoltaics: Fundamentals, Technologies and Applications.
3. Mukund R. Patel, Wind and Solar Power Systems.
4. Janaka Ekanayake et al., Smart Grid: Technology and Applications.

Approved public renewable-power sources

- https://nptel.ac.in/courses/108107113
- https://nptel.ac.in/courses/108105066

These sources provide the verified study basis while official Course 5033D detail is unavailable; they do not replace official power-plant designs, authorised grid-connection plans, certified equipment specs or authorised energy audits.